

Abhraneel Sarma

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Employment

2025- **Postdoctoral Researcher**
Institute of Human-Centered Computing, Graz University of Technology
Advisor: Alex Lex

Education

2019-2025 **PhD, Computer Science**
Northwestern University
Advisors: Matthew Kay, Jessica Hullman

2019-2021 **Master of Science, Computer Science**
Northwestern University

2016-2018 **Master of Science, Information**
University of Michigan
Advisor: Matthew Kay

2012-2016 **Bachelors, Design**
Indian Institute of Technology Guwahati

Awards and Honors

Papers

2024 **Best Paper Honorable Mention** for “Odds and Insights ...”, ACM CHI

2024 **Best Paper Honorable Mention** for “Opportunities, tensions ...”, ACM DIS

2023 **Best Paper Honorable Mention** for “multiverse ...”, ACM CHI

2022 **Best Paper Honorable Mention** for “Evaluating the use of ...”, IEEE VIS

2019 **Best Paper Award** for “Increasing the Transparency of ...”, ACM CHI

Other Activities

2021-2025 **Special Recognitions for Outstanding Reviews** (five), ACM CHI

Publications

Peer-reviewed Full-length Conference and Journal Articles

C12 2026 **Should We Dangle a Carrot? The Effect of Performance-based Incentives in Visualization Experiments**
[Abhraneel Sarma](#) , Matthew Kay, Sheng Long, Michael Correll and Alexander Lex
IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2026)

- C11 2026 **Cumulative Clarity: Using CCDF Plots for Visualizing Ordinal Regression Results**
[Abhraneel Sarma](#)
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2026)
- C10 2025 **More Forecasts, More (Decision) Problems: How Uncertainty Representations for Multiple Forecasts Impact Decision Making** [DOI](#)
[Abhraneel Sarma](#) , Maryam Hedayati and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
- C9 2024 **Odds and Insights: Decision Quality in Exploratory Data Analysis Under Uncertainty** [DOI](#)
[Abhraneel Sarma](#) , Xiaoying Pu, Yuan Cui, Eli T Brown, Michael Correll and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Honorable Mention
- C8 2024 **Milliways: Taming Multiverses through Principled Evaluation of Data Analysis Paths** [DOI](#)
[Abhraneel Sarma](#) , Kyle Hwang, Jessica Hullman and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
- C7 2024 **Opportunities, tensions, and challenges in computational approaches to addressing on-line harassment** [DOI](#)
 Evey Huang, [Abhraneel Sarma](#) , Sohyeon Hwang, Eshwar Chandrasekharan and Stevie Chancellor
 Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS)
Best Paper Honorable Mention
- C6 2024 **The multiverse of universes: A tutorial to plan, execute, and interpret multiverses analyses using the R package multiverse** [DOI](#)
 Martin Götz, [Abhraneel Sarma](#) and Ernest O’Boyle
 International Journal of Psychology
- C5 2023 **multiverse: Multiplexing Alternative Data Analyses in R Notebooks** [DOI](#)
[Abhraneel Sarma](#) , Alex Kale, Michael Moon, Nathan Taback, Fanny Chevalier, Jessica Hullman and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Honorable Mention
- C4 2022 **Evaluating the Use of Uncertainty Visualisations for Imputations of Data Missing At Random in Scatterplots** [DOI](#)
[Abhraneel Sarma](#) , Shunan Guo, Jane Hoffswell, Ryan Rossi, Fan Du, Eunye Koh and Matthew Kay
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2022)
Best Paper Honorable Mention
- C3 2021 **An automated approach to reasoning about task-oriented insights in responsive visualization** [DOI](#)
 Hyeok Kim, Ryan Rossi, [Abhraneel Sarma](#) , Dominick Moritz and Jessica Hullman
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2021)
- C2 2020 **Prior Setting in Practice: Strategies and Rationales Used in Choosing Prior Distributions for Bayesian Analysis** [DOI](#)
[Abhraneel Sarma](#) and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)

- C1 2019 **Increasing the Transparency of Research Papers with Explorable Multiverse Analyses** DOI
 Pierre Dragicevic, Yvonne Jansen, Abhraneel Sarma, Matthew Kay and Fanny Chevalier
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Award

Workshops and Lightly-Reviewed Articles

- W2 2024 **Tasks and Telephones: Threats to Experimental Validity due to Misunderstandings of Visualisation Tasks and Strategies**
Abhraneel Sarma, Sheng Long, Michael Correll and Matthew Kay
 evaluation and BEyond - methodoLogIcal approaches for Visualization (BELIV) Workshop at IEEE VIS
- W1 2019 **Interactive Visualizations Tools for Prior Setting In Bayesian Analysis: Challenges For Evaluation**
Abhraneel Sarma and Matthew Kay
 Workshop on Human-Centered Study of Data Science Work Practices at ACM CHI

Teaching

Instructor of Record

- 2025 **CS333 Interactive Information Visualization**
 Northwestern University

Conference Courses

- 2022, 2023 **Transparent Practices for Quantitative Empirical Research**
 Co-instructors: Chat Wacharamanotham, Fumeng Yang, Xiaoying Pu, Lace Padilla, Maryam Hedayati (this is a lecture series with rolling instructors).
 ACM CHI 2022 (online)
 ACM CHI 2023 (hybrid)
 IEEE VIS 2023 (in-person)

Teaching Assistantship

- 2020 **JOUR377 Data Analysis and Visualization**
 Northwestern University
- 2018 **SI330 Data Manipulation in Python**
 University of Michigan
- 2017 **SI588 Fundamentals of Human Behavior**
 University of Michigan

Mentoring

Northwestern University

- 2024-2026 **Yong-yu Huang**, undergraduate student for “Improving Trust by Communicating Uncertainty Due to Non-Response in Election Polls”
- 2024 **Sheng Long**, PhD student, co-mentored with Matthew Kay for “Tasks and Telephones...”
- 2024 **Mandi Cai**, PhD student, informal research mentoring
- 2021-2023 **Kyle Hwang**, undergraduate student for “Milliways ...”
- 2022 **Philip Clement**, undergraduate student for “Milliways ...”

Service

Program Committee

- 2026 **Program Committee Member, IEEE VIS**
- 2026 **Program Committee Member (Short Papers), EuroVis**
- 2024- **Journal of Visualization and Interaction, Open Practices Chair**

Paper Reviews

- 2024-2025 **The ACM Symposium on User Interface Software and Technology (UIST)**
- 2024-2025 **Information Visualization**
- 2023-2025 **IEEE Transactions on Visualization and Computer Graphics (TVCG)**
- 2022 **Workshop on TRust and EXpertise in Visualization (TREX VIS)**
- 2021-2023 **IEEE Visualization and Visual Analytics Conference (VIS)**
- 2021-2025 **The ACM Conference on Human Factors in Computing Systems (CHI)**

At Northwestern University

- 2022 **Computer Science Incoming PhD Student Orientation, Student Coordinator**
- 2020-2023 **Computer Science Social Initiative, Organising Committee**

Student Volunteering

- 2017-18 **IEEE Visualization and Visual Analytics Conference (VIS)**

Talks & Panels

Invited Talks

- 2025 **Designing Interfaces to Support Reliable Data-driven Decision-making**
Department of Computer Science, Washington University in St. Louis
- 2021 **Using the multiverse R package for Sensitivity Analysis**
Department of Engineering Sciences and Applied Mathematics, Northwestern University
- 2019 **Statistics and Scientific reporting: Can we do better?**
ICOS Big Data Summer Camp, University of Michigan

Discussion Panels

- 2024 **"Navigating Graduate School" Panel**
with Federico Sossai, Radhika Garg, Shubham Shahi and Ayse Hunt
CS401: Introduction to Graduate Studies, Northwestern University
- 2024 **PhD Panel**
with Dietrich Geisler, Melissa Chen and Achala Mishra
CS298: Computer Science Research Track Program, Northwestern University
- 2024 **Student Research & FAQ Panel**
with Maryam Hedayati, Tommy McMichen, Donna Hooshmand, Aidan Fitzsimons, Maryam Azmandian and Nathaniel Hejduk
Computer Science Incoming PhD Student Orientation, Northwestern University

Research Assistantships & Internships

- 2019-2025 **Graduate Research Assistant**
MUCollective, Northwestern University
- 2023 **Research Intern**
Microsoft Corp.
Mentors: Jake Hofman, Dan Goldstein
- 2021 **Research Intern**
Adobe, Inc
Mentor: Shunan Guo
- 2016-2019 **Graduate Research Assistant (full time, May 2018 onwards)**
MUCollective, University of Michigan
- 2017 **User Experience Intern**
Office of Academic Innovation, University of Michigan
- 2015 **Undergraduate Research Intern**
Keio-NUS CUTE Center, National University of Singapore